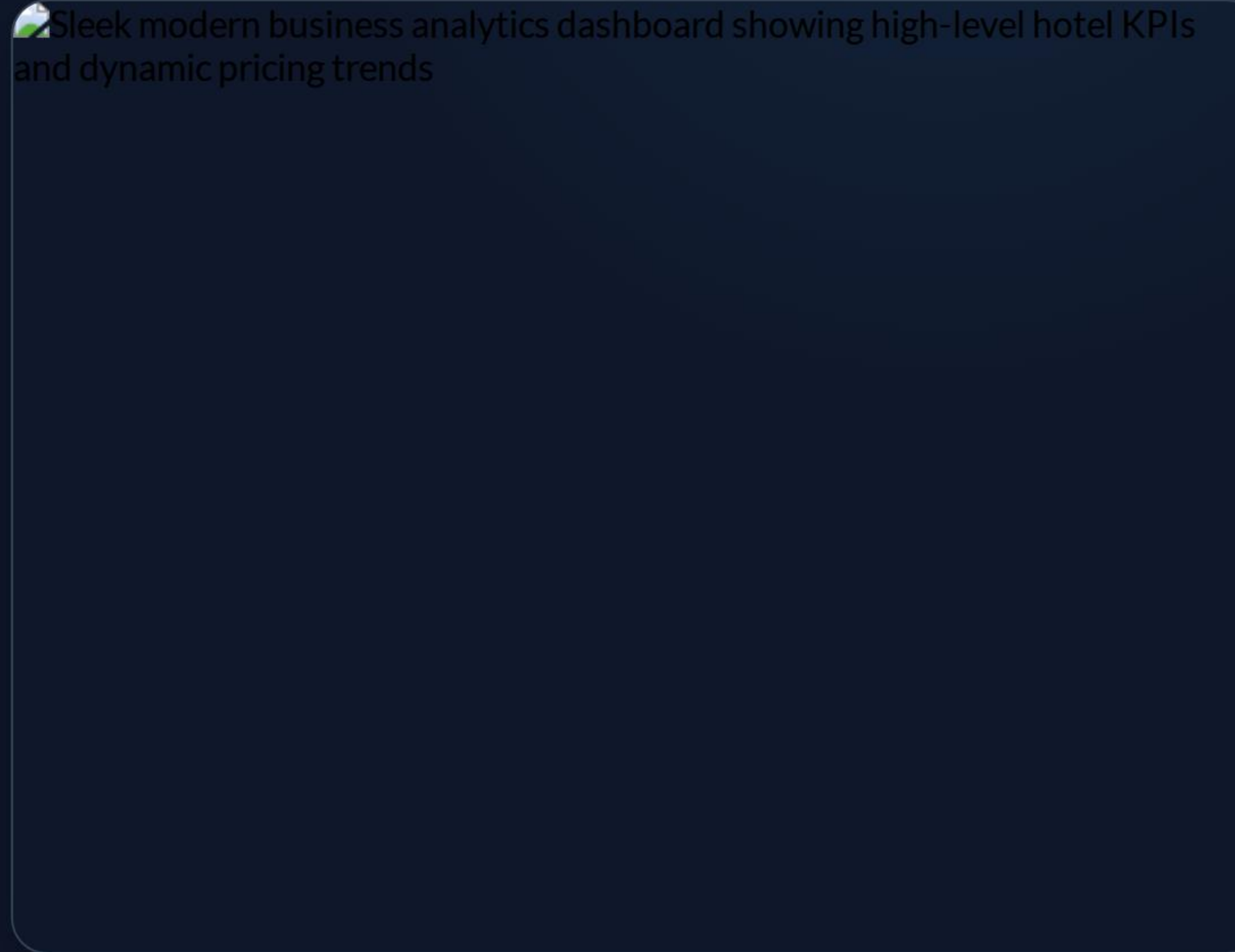


Travel, Tourism & Hospitality **Data Analytics**

Mid-Project Review: Customer Retention & Dynamic Pricing Analysis

Project Overview

- **Core Objective:** Identify the definitive historical and behavioral drivers behind hotel reservation cancellations.
- **Predictive Framework:** Build a robust, machine-learning-backed model to accurately forecast at-risk bookings.
- **Business Value:** Enable management to optimize dynamic pricing strategies, mitigate churn, and significantly reduce revenue loss.



Sleek modern business analytics dashboard showing high-level hotel KPIs and dynamic pricing trends

Phases 1 & 2: Data Engineering & EDA



Price Sensitivity

A strong positive correlation exists between Average Daily Rate (ADR) and churn. Higher prices directly increase the likelihood of price-shopping and cancellations.



Channel Risk

Reservations originating from Online Travel Agencies (OTAs) exhibit a drastically higher cancellation rate compared to direct or corporate bookings.



Lead Time Danger

Bookings made well in advance carry a significantly higher churn probability than last-minute reservations, requiring proactive engagement.

Phase 3: Predictive Modeling



77%

Cancellation Recall Rate

Transitioned from historical analysis to a robust predictive model, engineering new time-series metrics while actively removing retrospective data-leakage columns.

We overcame severe target class imbalance by utilizing Standard Scaling and applying balanced class weights. The resulting Logistic Regression classifier achieved exceptional accuracy in identifying at-risk bookings.

Phase 4: Business Intelligence

We successfully exported the cleaned, scored dataset and established a robust **Star Schema** architecture within Power BI.

Our central Fact table is fully operational and strictly mapped to contextual dimension tables (Hotel Type, Market Segment). Initial DAX measures are tracking core executive KPIs.

All backend scripts and visualizations are secured via Git version control on our dedicated development branch.



Status & Next Steps



Current Project Status

We are currently operating ahead of schedule with zero technical blockers. The foundational data architecture is complete.



Visual Dashboard Deployment

Focusing heavily on the UI/UX front-end build. We will be integrating interactive slicers and clustered bar charts to visualize churn by market segment.



Final Reporting & Delivery

Upon dashboard completion, we will merge the development branch and compile the final data-driven business recommendation report.

Questions?

Thank you for your time. Happy to discuss the methodology, the machine learning model, or our Power BI architecture.

Image Sources



<https://static.vecteezy.com/system/resources/thumbnails/049/191/806/small/a-sleek-dashboard-displays-dynamic-analytics-and-performance-indicators-showcasing-a-modern-interface-for-enterprise-resource-planning-photo.jpg>

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